

Measuring Robustness of Pruned Models

Robustness Transfer at Initialization

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The question. Pruning at Initialization (PaI) finds sparse subnetworks *before* training, which makes the mask a durable artifact that is computed once on one dataset and reused on another (transfer/federated learning, rapid prototyping). Two literatures left a precise gap: PaI papers report only clean accuracy, and adversarial-pruning papers only prune *after* training. **Does the adversarial robustness of a PaI mask survive a change of dataset?** This work answers it for the CIFAR-10 $\rightarrow$ CIFAR-100 transfer.

Novelty

- **Two new metrics (RTI & nRTI).** The Robustness Transfer Index extends the Data Transfer Index from clean to *adversarial* accuracy. Because absolute-gap RTI collapses under strong attacks (both accuracies fall to the robustness floor), I introduce the **normalized nRTI**, a floor-corrected relative metric that stays discriminative from FGSM through AutoAttack.
- **First systematic study of the empty cell:** $\text{PaI} \times \text{adversarial accuracy} \times \text{cross-dataset transfer}$ — a setting no prior PaI or adversarial-pruning work had evaluated.
- **Comprehensive benchmark:** 5 PaI methods (SNIP, GraSP, SynFlow, DPaI, YOPO) $\times$ 3 architectures (VGG-19, ResNet-18, ResNet-20) $\times$ 7 sparsities $\times$ 5 attacks = 525 evaluation cells, all under the ℓ_∞ threat model and reported AutoAttack-credibly, not PGD-only.
- **A falsifiable structural claim:** how data-dependent a method’s saliency criterion is mechanically predicts how well its robust structure transfers — recasting PaI as a *discoverer of transferable robust structure*, not merely a compression tool.

Key Results

- **Data-independence predicts transfer ($r \approx 0.97$).** The degree of data-dependence of the saliency criterion correlates almost perfectly with nRTI. Fully data-free methods (SynFlow, YOPO) reproduce native robustness with high fidelity; data-heavy methods do not.
- **nRTI is necessary.** From FGSM to AutoAttack, RTI compresses $\sim 13\times$ ($1.43 \rightarrow 0.11$) and would falsely rank collapsed methods as “best”; nRTI compresses $< 3\times$ and preserves the ranking.
- **Residual architectures amplify the advantage.** ResNet-20 gives the lowest nRTI for *every* method; data-dependent saliency interacts worst with plain VGG-19.
- **A 50–68% sparsity sweet spot** where transferred subnetworks match or beat natively-pruned ones.
- **Topology carries robustness.** For SynFlow and YOPO on residual nets, the transferred mask matches the native mask to within test-set noise — evidence that robustness is encoded, at least partly, in the mask *topology* itself.

PaI method	Data-dependence	Mean nRTI (%)	Transfer
YOPO	data-free (level 0)	5.25	best
SynFlow	synthetic (level 1)	6.18	strong
SNIP	gradient (level 2)	10.00	moderate
GraSP	Hessian (level 2)	10.61	moderate
DPaI	iterative (level 3)	13.52	weakest

Lower nRTI = more faithful cross-dataset robustness transfer. Ordered along the data-dependence axis.

Takeaway: Cross-dataset transferability of adversarial robustness in PaI subnetworks is governed — to a degree that is empirically substantial ($r \approx 0.97$) — by the *data-independence* of the saliency operator that produced the mask, and is further amplified by residual connections.